

BodyShield Full Paper Draft (Analytic/Simulation Scope)

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Abstract

BodyShield is presented here as an analytic/simulation study of falsifying and repairing hidden embodiment-control assumptions. The package contains simulation, bounded high-fidelity probes, local surrogate policy repair summaries, audits, and release verification. Hardware and external checkpoint evidence are explicitly blocked.

Method

BodyBreak searches a finite perturbation set for low-cost policy-breaking embodiment-control shifts. BodyShield repairs against discovered break axes and evaluates nominal retention plus held-out perturbation families. The current method validates a falsification-to-repair workflow, not deployment transfer.

Evidence

Completed tiers include analytic simulation, search/repair, claim ledger, citation verification, local release bundle, bounded high-fidelity probes, and reviewer-risk reports. Partial tiers include local policy surrogates and analytic oracle feasibility.

Blockers

PAPER NOT READY: hardware validation/noise floor/verifier/reset/physical modifications/videos are not run; external trained-policy checkpoints and full-scale rollouts are missing; real-video WAM and corrective-trace datasets are missing; oracle feasibility is analytic only; BodyBreak minimality is estimated rather than globally proven; release is local rather than independently archived; human paper review remains open.

Allowed Wording

Use analytic/synthetic scope. Do not claim hardware validation, real-video WAM results, real corrective-trace adaptation, external trained-policy benchmark closure, global minimality, autonomous lab operation, or broad superiority over domain randomization, robust control, sysID, or CBF methods.